Understanding the Cost of Compression: BERT vs. DistilBERT

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GitHub repo: https://github.com/JodieeT/DistilBERT

1 Introduction

Large pretrained models like BERT [1] achieve high downstream accuracy but carry costs that are prohibitive in resource-constrained settings. Sanh et al. [4] propose DistilBERT, a 6-layer student trained against the 12-layer teacher via knowledge distillation, and report that it is **40% smaller**, **60% faster**, and retains **∼97%** of BERT's GLUE accuracy. Our research question is not whether those numbers replicate (they largely do) but *where the lost 3% lives*. The headline 97% obscures task-dependent failure modes; we extend the comparison to three datasets stressing different linguistic phenomena, then read per-sample disagreements between teacher and student to identify what kinds of inputs the student fails on.

2 Chosen Result

We reproduce Sanh et al.'s Table 2 (the 40%/60%/97% claim on GLUE), the paper's central empirical claim. We chose it because the headline retention is suspiciously *aggregate* — a single number across heterogeneous tasks tells you nothing about which input distributions break first.

3 Methodology

Models. `bert-base-uncased` (12 layers, 110M params) vs. `distilbert-base-uncased` (6 layers, 67M params), both via HuggingFace `transformers` [6]. We do not redistil; we use the public DistilBERT checkpoint.

Datasets. (i) SST-2 [5]: 67K train / 872 val, single-sentence sentiment. (ii) IMDb [3]: 25K / 25K, multi-paragraph reviews (∼230 words median). (iii) MRPC [2]: 3.7K / 408, sentence-pair paraphrase identification with pair tokenization `tokenizer(s1, s2)`.

Fine-tuning. 3 epochs, AdamW, lr 2×10^{-5}, weight decay 0.01, batch size 16 (IMDb) / 32 (SST-2/MRPC), max length 128 / 256, on a single Colab T4 GPU.

Evaluation. Accuracy, F1, training wall-clock, parameters, and inference latency computed as 1/`eval_samples_per_second` from `trainer.evaluate()` — one consistent throughput methodology across datasets.

Beyond the paper. We dump per-sample BERT and DistilBERT predictions, join them, and label each row with one of four categories (`both_correct`, `both_wrong`, `bert_only_correct`, `distilbert_only_correct`); then bucket by length, flag negation/contrast, compute confidence calibration, and hand-curate qualitative case studies on the disagreements.

4 Results and Analysis

Headline replication. Table 1 confirms all three of the paper's claims: DistilBERT has 39% fewer params (paper: 40%), trains 30–48% faster (paper: 60%; the smaller speed-up is consistent with input-pipe and tokenization overhead dominating wall-clock at our batch sizes on T4), and retains 96.9% / 99.1% / **101.8%** of BERT's accuracy on the three tasks (paper: ∼97%).

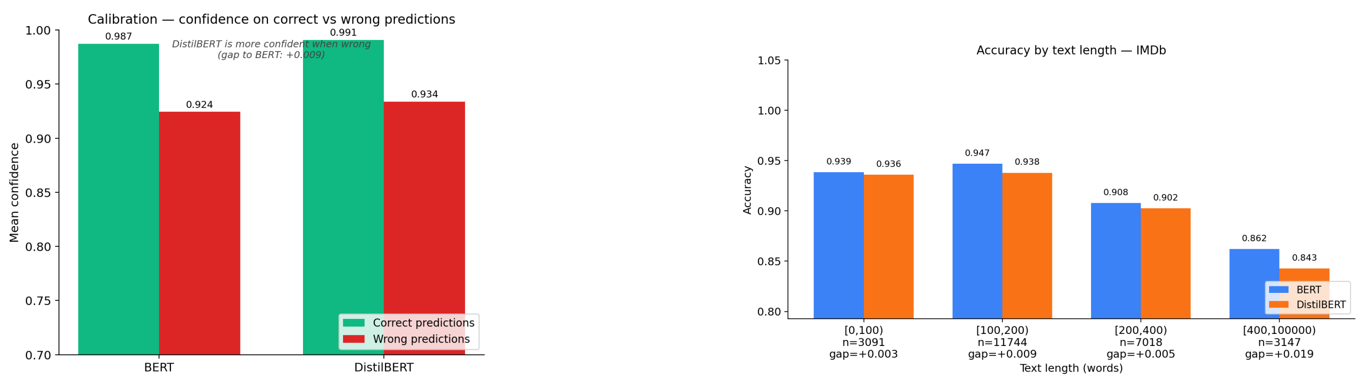
Where the gap lives. The headline ("97%") hides three sharply different stories. *SST-2* (−2.87 pp): the student fails on **compositional** sentiment — negation (*"i don't think i laughed out loud once"*), concessive structure (*X but Y*, where the verdict sits in clause Y), sarcasm, and idiomatic positives (*almost unsurpassed*,

	SST-2 (n=872)		**IMDb** (n=25,000)		**MRPC** (n=408)	
	BERT	DistilBERT	BERT	DistilBERT	BERT	DistilBERT
Accuracy	**0.928**	0.899	**0.924**	0.916	0.821	**0.836**
F1	**0.930**	0.902	**0.925**	0.916	0.876	**0.883**
Accuracy gap (pp)	−2.87		−0.84		**+1.47**	
Training time (s)	391	275	1013	530	52	31
Training speed-up	1.42×		1.91×		1.65×	
Parameters	109.5M	67.0M	109.5M	67.0M	109.5M	67.0M
Agreement rate	95.3%		95.2%		86.3%	

Table 1: BERT vs. DistilBERT across three GLUE-style benchmarks. Bold marks the winner.

not the least of which); the gap doubles on negated sentences (4.3 pp vs. 2.0 pp). *IMDb* (−0.84 pp): the aggregate gap is small but *grows with document length* (0.3 pp under 100 words → **1.9 pp** over 400 words; Fig. 1b); the dominant failure mode is *buried verdict* — a long descriptive review whose true polarity surfaces only in the last paragraph. *MRPC* (+1.47 pp, **student wins**): on a small training set (3.7K pairs), BERT's extra capacity becomes a liability — 71% of `distilbert_only_correct` cases are non-paraphrases that BERT incorrectly classified as paraphrases on shared surface material. Reduced capacity acts as a regulariser.

Calibration is its own failure mode. On SST-2, DistilBERT's mean confidence on its 33 distillation-cost mistakes is **0.893**, despite 0% accuracy on those samples (Fig. 1a); the same pattern holds on IMDb (0.867 on 700 mistakes). The student does not signal uncertainty when wrong, so confidence-thresholding is not a useful safety net.



(a) SST-2: DistilBERT's mean confidence on its mistakes (∼0.89) is barely below its confidence on its successes.

(b) IMDb: the teacher–student gap grows with review length, consistent with long-context integration as a failure mode.

Figure 1: Diagnostic figures characterising where DistilBERT loses to BERT.

5 Reflections

Lessons. The paper's single retention number obscures task-dependent failure modes; reading per-sample disagreements was far more informative than additional aggregate metrics. Compositional structure (negation, contrast, sarcasm) is where capacity matters — bag-of-tokens cues survive distillation, syntax less so. The MRPC inversion was the most surprising result: on small datasets the student can be the safer choice precisely *because* of its reduced capacity. Calibration is independent of accuracy: DistilBERT is "confidently wrong," so confidence thresholding is not a safety net.

Challenges. The paper's claimed 60% training speed-up was unreachable on a single T4: input-pipe and tokenization overhead capped ours near 48%.

Future directions. Train a calibration-aware student with temperature scaling or label smoothing inside the distillation loss; explore robustness distillation via contrastive or saliency-matching objectives targeting the negation/sarcasm cases identified here; test whether these failures generalise to other compressed encoders (TinyBERT, MobileBERT).

References

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